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Alkylsulfate-based ionic liquids to separate azeotropic mixtures☆

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ABSTRACT

The knowledge of liquid–liquid equilibria (LLE) of the ternary systems (hexane or heptane + ethanol + 1-ethyl-3-methylimidazolium ethylsulfate (EMIM $EtSO_4$)) is essential for the separation of alkanes from their azeotropic mixtures with ethanol. The experimental LLE have been determined at 298.15 K and atmospheric pressure. Experimental LLE are correlated using NRTL equation. The solvent capacity of the IL is compared with others aiming to analyze the efficiency of these molten salts used as entrainers. The extraction processes with this IL are simulated using conventional software. A comparison of the alkylsulfate-based IL's ability for the extraction process, determined from the simulation results, is enclosed.

1. Introduction

Gasoline is a mixture of hydrocarbons with a high tendency for knocking. To avoid this problem and consequently raise the octane percentage as close as possible to 100, anti-knock compounds such as tetraethyl and tetramethyl lead have been extensively used in the past. Although these products have proved to have the best performance with regard to their cost/effectiveness ratio, they have generated very important atmospheric pollution problems.

Since the 90s, the legislation of most industrialized countries has envisaged a greater reduction of lead levels in gasoline. The manufacture of oxygenated additives has emerged as a safety alternative to the conventional lead addition, albeit several azeotropes such as hexane or heptane with ethanol are present. Taking into account the abovementioned, the foundations of this work are placed in the problems that are faced in petrochemical industries, when the separation of azeotropic mixtures is required [1]. One of the most common ways to separate this kind of mixtures is through extractive distillation [2]. Nevertheless, the high-energy requirements to get a fluid phase system, the existence of volatile organic compounds and high pressures, are major drawbacks [3] that point out liquid–liquid extraction as a more environmentally friendly separation process.

Up to date, one of the most important challenges on chemical engineering is the development of greener and more efficient processes. The search for something new in terms of chemical engineering, has triggered a lot of scientific and commercial interests in ionic liquids (ILs). These molten salts have been suggested as a key alternative to be employed to separate liquid mixtures. The huge number of possible combination of cations and anions allows the design of task specific ILs, suitable for the final objectives of the process [4]. Besides that, the very low volatility at room-temperature [5], non-flammability [6] and recyclability [7] gives them the label of greener than common organic solvents.

More specifically, ILs demonstrated to possess the ability to act as extraction solvents to separate azeotropic mixtures, including outstanding examples such as ethanol + water [8–10], THF + water [8,9,11], alcohol + alkane mixtures [12–16], aromatic + aliphatic mixtures [17,18], ethylacetate + alcohol [19,20], ethylacetate + hexane [21], ketone + alkane or alcohol mixtures [22,23] and ethyl tert-butyl ether + ethanol [24].

In this investigation, the selection of the IL extraction solvent was built on the basis of an economical and efficient point of view. The selected alkylsulfate-based ILs with a cation derived from imidazolium [25] were considered as some of the most promising ILs to be applied in industrial processes. In general, they can be easily synthesized in an atom-efficient and halide-free way, at a reasonable cost. They show chemical and thermal stabilities, low melting points and relatively low viscosities. Particularly, 1-ethyl-3-methylimidazolium ethylsulfate (EMIM $EtSO_4$) combines those features in a suitable way to be used as solvent in liquid–liquid extraction.

In this work, ternary LLE of hexane or heptane + ethanol + EMIM $EtSO_4$ are determined at 298.15 K and atmospheric pressure, as the knowledge of the (alkane + ethanol + IL) LLE is essential for the design of the separation technique applied, in terms of the number of ideal stages or the most appropriate solvent. Additionally, the capacity of ILs-like solvents in liquid extraction processes is

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Table 1
Comparison of density (ρ) and refractive index (n_D) of pure components at 298.15 K with the literature data.

Component	ρ (g cm^{-3})		n_D	
	Exp.	Lit.	Exp.	Lit.
Hexane	0.6549	0.65484 [32]	1.37217	1.37226 [32]
Heptane	0.6795	0.67946 [32]	1.38517	1.38511 [32]
Ethanol	0.7851	0.78517 [33]	1.35929	1.35924 [33]
EMIM EtSO$_4$	1.2375	1.23763 [34]	1.47940	1.47940 [34]

investigated using the selectivity and the solute distribution ratio. Preliminary studies of activity coefficients at infinite dilution of the methylimidazolium-based ILs have in their favour the low activity coefficients of ethanol and high ones of heptane [26–30]. As these results suggest, these ILs are selected as solvent in the case of the mixture heptane with ethanol. Moreover, the experimental data are correlated by applying the NRTL equation [31], thus rendering possible their implementation and use in the simulation of the extraction process by means of the HYSYS software. The simulation results will permit to test the quality of the IL as an azeotrope breaker.

2. Experimental

2.1. Chemicals

Gas chromatography of hexane (from Fluka, ≥99 wt%), heptane (from Aldrich, ≥99.0 wt%) and ethanol (from Merck, ≥99.8 wt%) verified their nominal purities. EMIM EtSO$_4$ was purchased from Solvent Innovation with a purity ≥99 wt%.

Vacuum (10^{-1} Pa) and moderate temperature (323 K) conditions were always applied to the IL for several days prior to its use, in order to reduce the water and other volatile substances content. NMR of the IL was determined to ensure their purity. The final water mass fraction was measured by Karl Fisher coulometric titration (Metrohm 831 KF Coulometer) and the dried liquid contained 362 ppm.

The density and the refractive index of the pure chemicals together with the recent literature values [32–34] are presented in Table 1.

2.2. Apparatus and experimental LLE procedure

A glass cell was used in order to determine the binodal curve and the LLE tie-lines that show the composition of each layer. Ternary mixtures of known composition were placed into the cell and thermostatted by a water jacket connected to a bath controlled to ±0.01 K. The temperature in the cell was measured with an ASL F200 digital thermometer with an uncertainty of ±0.01 K. Each mixture was stirred vigorously for 1 h and left to settle for 4 h to guarantee a complete split of the equilibrium phases. Then, samples of both layers were taken with a syringe for composition analysis by means the densities and refractive indices. The compositions were inferred by using calibration curves which had previously been constructed at 298.15 K. These curves were obtained through fitting the composition on the binodal curve by means of densities and refractive indices at 298.15 K. The uncertainty of the phase composition is estimated as $\pm 4 \times 10^{-3}$ in mass fraction.

All weighing were carried out in a Mettler AX-205 Delta Range balance with an uncertainty of $\pm 10^{-4}$ mass fraction. Densities were measured with an Anton Paar DSA-48 digital vibrating tube densimeter with an uncertainty of $\pm 2 \times 10^{-4}$ g cm^{-3}, and refractive indices with a Dr. Kernchen ABBEMAT WR automatic refractometer with an uncertainty of $\pm 4 \times 10^{-5}$. The apparatus was calibrated

Table 2
Composition of the experimental tie-lines, solute distribution ratio (β) and selectivity (S) for the ternary systems at 298.15 K.

Alkane-rich phase		IL-rich phase		β	S
x_1^{HC}	x_2^{HC}	x_1^{IL}	x_2^{IL}		
Hexane (1) + ethanol (2) + EMIM EtSO$_4$ (3)					
1.000	0.000	0.000	0.200	n.a.	n.a.
0.999	0.001	0.000	0.332	n.a.	n.a.
0.995	0.005	0.000	0.452	n.a.	n.a.
0.989	0.012	0.013	0.541	47.02	3631.0
0.977	0.023	0.025	0.602	26.52	1020.4
0.935	0.065	0.046	0.681	10.50	215.37
0.884	0.116	0.057	0.740	6.37	98.45
0.815	0.185	0.090	0.767	4.14	37.68
0.787	0.208	0.157	0.746	3.58	17.93
0.673	0.317	0.237	0.709	2.24	6.35
0.652	0.338	0.299	0.658	1.95	4.24
Heptane (1) + ethanol (2) + EMIM EtSO$_4$ (3)					
0.988	0.012	0.000	0.142	11.85	n. a.
0.985	0.015	0.000	0.297	19.27	n. a.
0.980	0.020	0.001	0.434	22.15	n. a.
0.975	0.025	0.009	0.537	21.16	2343.2
0.971	0.029	0.014	0.608	20.98	1414.4
0.937	0.063	0.029	0.697	10.99	358.67
0.877	0.123	0.050	0.748	6.06	106.00
0.826	0.174	0.072	0.777	4.46	50.91
0.761	0.239	0.099	0.782	3.27	25.20
0.675	0.325	0.130	0.779	2.40	12.41
0.616	0.383	0.191	0.750	1.96	6.32
0.509	0.490	0.249	0.709	1.45	2.96

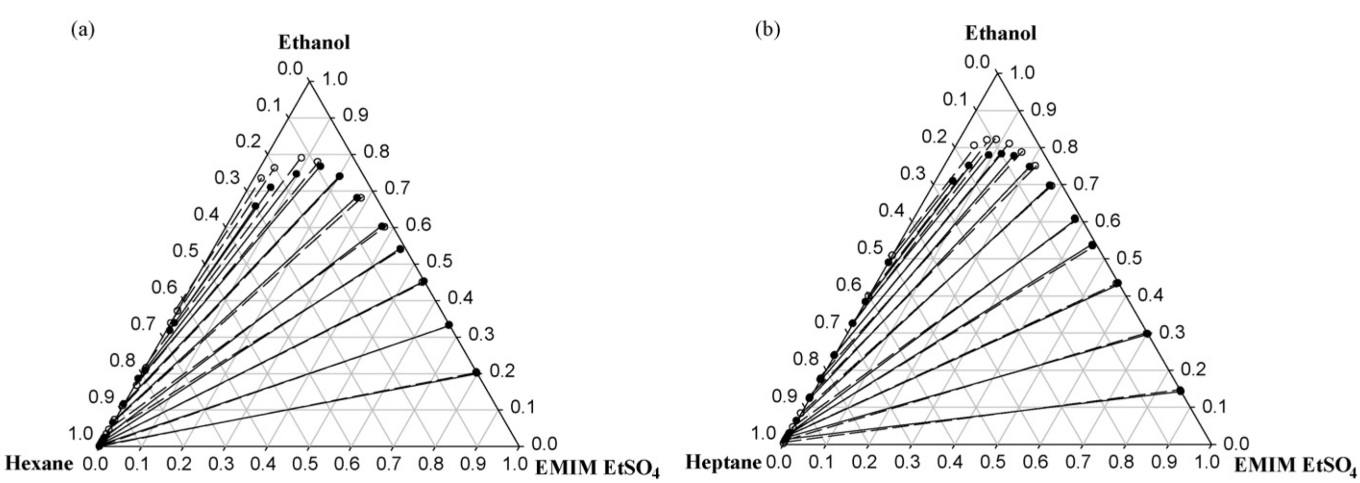

Fig. 1. Experimental tie-lines of the ternary systems: (a) hexane (1) + ethanol (2) + EMIM EtSO$_4$ (3); (b) heptane (1) + ethanol (2) + EMIM EtSO$_4$ (3) at 298.15 K.

Table 3
Fitting parameters and standard deviation of the LLE data by means of the NRTL equation.

System	i, j	Δg_{ij} (J mol^{-1})	Δg_{ji} (J mol^{-1})	α	σ
Hexane (1) + ethanol	1, 2	3307.6	1972.6	0.10	0.022
(2) + EMIM $EtSO_4$ (3)	1, 3	49311	1121108		
	2, 3	−7412.8	−11159		
Heptane (1) + ethanol	1, 2	2323.8	2123.1	0.16	0.023
(2) + eMIM $EtSO_4$ (3)	1, 3	29645	19265		
	2, 3	−12162	9602.9		

Table 4
Properties of the main streams in the extraction of ethanol from its azeotropic mixture with hexane using EMIM $EtSO_4$ as solvent by means of HYSYS software.

Stream name	Solvent	Feed	Raffinate	Extract	Ethanol
Temperature (K)	298.15	298.15	298.15	298.15	298.15
Pressure (kPa)	101.32	101.32	101.32	101.32	101.32
Molar flow (g mol/h)	3.93	15.47	11.28	8.12	4.19
Mass flow (kg/h)	0.5091	1.1282	0.9374	0.6999	0.1908
Liquid volume flow (mL/h)	457.0	1643.0	1381.1	718.9	261.9
Mass fraction (w)					
Hexane	0	0.7916	0.9085	0.0592	0.2172
Ethanol	0.2000	0.2084	0.0635	0.3964	0.7828
EMIM $EtSO_4$	0.8000	0	0.0280	0.5444	0

with Millipore quality water and ambient air for the densimeter, and with Millipore quality water and tetrachloroethylene (supplied by the company) for the refractometer, according to the manufacturer instructions.

3. Results and discussion

LLE data for the ternary mixtures (hexane or heptane + ethanol + EMIM $EtSO_4$) at 298.15 K and atmospheric pressure are reported in Table 2. Triangular diagrams with the graphical representation of the ternary LLE data are shown in Fig. 1, where the IL is concluded to be completely immiscible in the alkane. The immiscibility region is practically independent of the size of the hydrocarbon and the slope of the tie-lines point out towards obtaining a high purity of the alkane to be removed from the azeotropic mixture using EMIM $EtSO_4$ as extraction solvent. These data make up a promising onset for the implementation of a successful scaling up with industrial application.

3.1. Solute distribution ratio and selectivity

The solute distribution ratio (β) provides the solvent capacity of the IL, related to the amount of the solvent required for the extraction process. The selectivity (S) is an important parameter in assessing the efficiency of the desired IL for the effective extraction of ethanol from the azeotropic system based on previously obtained β data. Values of solute distribution ratio and selectivity are shown in Table 3. These parameters were calculated according to the equations:

$$\beta = \frac{x_2^{\text{IL-phase}}}{x_2^{\text{HC-phase}}} \tag{1}$$

$$S = \left(\frac{x_1^{\text{HC-phase}}}{x_1^{\text{IL-phase}}}\right)\left(\frac{x_2^{\text{IL-phase}}}{x_2^{\text{HC-phase}}}\right) \tag{2}$$

A comparison with other ILs [7,13–16] was made and no significant differences were recorded. High values of these parameters were obtained in all cases and then, they could be considered as potential good candidates for the removal of alkanes from their azeotropic mixtures through an extraction process. In this case, high selectivity values are obtained especially at hydrocarbon compositions close to the pure alkane.

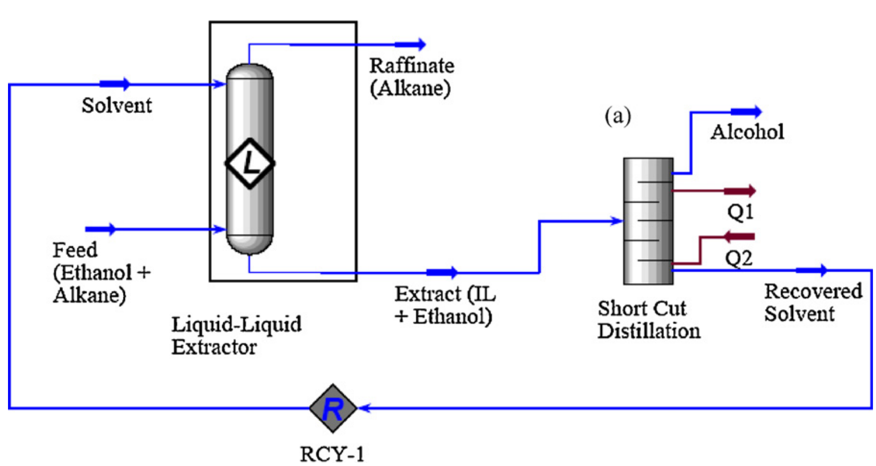

Fig. 2. Process flowsheet in the purification of alkanes from its azeotropic mixture with ethanol using EMIM $EtSO_4$.

Table 5
Properties of the main streams in the extraction of ethanol from its azeotropic mixture with heptane using EMIM $EtSO_4$ as solvent by means of HYSYS software.

Stream name	Solvent	Feed	Raffinate	Extract	Ethanol
Temperature (K)	298.15	298.15	298.15	298.15	298.15
Pressure (kPa)	101.32	101.32	101.32	101.32	101.32
Molar flow (g mol/h)	5.70	16.55	5.66	16.58	10.88
Mass flow (kg/h)	0.7373	1.058	0.5076	1.287	0.5497
Liquid volume flow (mL/h)	661.8	1438.2	728.9	1371.1	709.3
Mass fraction (w)					
Heptane	0	0.5172	0.8996	0.0702	0.1644
Ethanol	0.2000	0.4828	0.1003	0.4717	0.8356
EMIM $EtSO_4$	0.8000	0	0.0001	0.4581	0

Table 6
Comparison of simulation results of extraction process using alkylsulfate based ILs as azeotrope breaker in (heptane or hexane + ethanol + IL) ternary systems by means of HYSYS software.

Ternary system		Raffinate stream		
Azeotropic mixture	Solvent	Alkane stream	Ethanol stream	IL stream
Hexane + ethanol	MMIM $MeSO_4$	0.8986	0.0990	0.0024
	EMIM $EtSO_4$	0.9085	0.0635	0.0280
	BMIM $MeSO_4$	0.9777	0.0202	0.0021
Heptane + ethanol	MMIM $MeSO_4$	0.8845	0.1121	0.0034
	EMIM $EtSO_4$	0.8996	0.1003	0.0001
	BMIM $MeSO_4$	0.9437	0.0563	0

3.2. Correlation with NRTL model

The experimental tie-lines of the ternary mixtures were correlated in order to carry out a simulation study that will be a valuable tool which will render possible the implementation of the process at industrial scale. The classical non-random two liquid (NRTL) model was chosen [26], since it has already been used for appropriately correlating the LLE data of analogous ternary systems involving hydrocarbons and ILs [35–37].

For the NRTL model, the third randomness parameter α was optimized and their values are reported in Table 3. In this equation, the adjustable parameter was defined as follows:

$$\tau_{ji} = \frac{\Delta g_{ij}}{RT} \quad (3)$$

The parameters were adjusted to minimize the difference between the experimental and calculated mole fractions defined as:

$$\begin{aligned}\text{O.F.} = \sum_{i=1}^{n} & [(x_{1i}^{\text{HC-phase}} - x_{1i}^{\text{HC-phase}}(\text{calc}))^2 \\ & + (x_{2i}^{\text{HC-phase}} - x_{2i}^{\text{HC-phase}}(\text{calc}))^2] \\ & + \sum_{i=1}^{n}((x_{1i}^{\text{IL-phase}} - x_{1i}^{\text{IL-phase}}(\text{calc}))^2 \\ & + (x_{2i}^{\text{IL-phase}} - x_{2i}^{\text{IL-phase}}(\text{calc}))^2) \quad (4)\end{aligned}$$

The fitting parameters of the equation are listed in Table 4 together with the standard deviations. These deviations were calculated by applying the following expression:

$$\sigma = \left(\frac{\sum_i (x_{ilm}^{\text{exp.}} - x_{ilm}^{\text{calc.}})^2}{6k} \right)^{1/2} \quad (5)$$

A discussion of the standard deviation between the experimental data and those obtained from their correlation by the NRTL equation indicates that the selected model was appropriate to correlate the immiscible region of these mixtures.

3.3. Simulation results

The purification of alkane by means of an effective extraction process using ILs as the solvent is proposed in this work. The flowsheet of the system is made up of a countercurrent continuous liquid–liquid extraction column including a solvent recovery stage. This goal can be better achieved by the simulation tools provided by HYSYS (v3.2) software from Aspen Technology, Inc. (Cambridge, MA). The performance of the process shown schematically in Fig. 2, where a liquid–liquid extractor with one equilibrium stage models the extraction column and the solvent recovery is modelled by a short-cut distillation process. The operating conditions were optimized in the neighbourhood of those obtained from the theoretical ones.

In this paper, the operating conditions were chosen from a compromise between cost and alkane raffinate purity. A high purity of alkane in the raffinate stream needs a minimum mass solvent/feed ratio (S/F). If this last parameter is minimum, it would be possible to feed a high quantity of azeotropic mixture with a minimum IL consumption. A mass balance involving purity of the raffinate stream, azeotropic feed and several ionic liquid purities was carried out in order to afford raffinate purities higher than 90 wt% with a S/F of 0.4 and 0.7 for hexane or heptane azeotropic mixture, respectively. The recovered solvent stream purity that feed the extractor is 80 wt%.

The strategy for the development of the simulation process was based on the following considerations. The parameters of the obtained NRTL correlation at 298.15 K were used in order to obtain the equilibrium data. The selected operation unit was the liquid extractor, which was isothermal at 298.15 K and 101.32 kPa. The simulated azeotropic feed had a 79 and 52 wt% hexane or heptane azeotropic mixture, respectively. Solvent mass flow was varied in order to get a optimum S/F. In summary, the solvent and feed compositions were kept invariable and flow rates were optimized to maximize raffinate purity.

The solvent and feed flow rates are listed in Tables 4 and 5. A raffinate purity of 90 wt% was achieved in both cases studied. The distillation stage of the process proposed allows to obtain an alcohol stream containing 80 wt% of ethanol. In this case, the ability of EMIM $EtSO_4$ to remove the alkane from the azeotropic mixture is patently ascertained.

Furthermore, a comparison between the data obtained in this work with other extraction processes involving other sulfate anion-based ILs previously reported by our group [7,14–16], is included in Table 6. Generally speaking, employing this kind of ILs allows to reach purities better than 90 wt% in the alkane. In detail, the use of EMIM $EtSO_4$ as extraction solvent will entail the ability to treat high amount of feed stream with a minimum quantity of IL, what will be translated into a more efficient process from an economical point of view.

4. Conclusion

The general picture of this work is the design of a cleaner process based on liquid–liquid extraction that could replace the azeotropic or extractive distillation conventional ones by employing ILs as solvents. In this investigation, it has been demonstrated the extremely appropriate ability of the EMIM $EtSO_4$ as an extraction solvent, that allows to purify alkanes from their azeotropic mixtures with ethanol. The LLE data allow the identification of theoretically appropriate operating conditions, for a room-temperature countercurrent continuous extraction process including a solvent recycling stage; these conditions were optimized by simulation techniques. The optimum computerized results showed an alkane raffinate purity of about 90 wt% which make up a feasible basis of an implementation of this process at laboratory scale. In summary, the process proposed here is preferred due to economical reasons, because the energy requirements for the extraction with the IL as solvent are much lower than those entailing distillation.

List of symbols

n_D refractive index
x mole fraction
w mass fraction
S selectivity
Δg_{ij} fitting parameters of NRTL model (J mol^{-1})
R ideal gas constant (J mol^{-1} K)
T temperature (K)
$x_{1i}^{\text{HC-phase}}$, $x_{2i}^{\text{HC-phase}}$, $x_{1i}^{\text{IL-phase}}$, $x_{2i}^{\text{IL-phase}}$ experimental mole fractions
$x_{1i}^{\text{HC-phase}}$(calc), $x_{2i}^{\text{HC-phase}}$(calc), $x_{1i}^{\text{IL-phase}}$(calc), $x_{2i}^{\text{IL-phase}}$(calc) calculated mole fractions
k number of experimental tie-lines
S/F solvent-to-feed ratio

Greek letters

ρ density (g/cm^3)
β solute distribution ratio
τ_{ij} parameters of NRTL equation
α parameter of NRTL model
σ standard deviation

Superscripts

calc., exp. calculated and experimental values
HC-phase, IL-phase hydrocarbon and ionic liquid rich phases, respectively

Subscripts

1 alkane (inert)
2 ethanol (solute)
3 IL (solvent)
i component
i, j parameter of NRTL model
m tie-lines
l phase

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